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Research question formulation: The goal of our project is to determine differences in gene expression between normal and diseased liver tissues, and use these insights to develop a method for predicting diagnosis of liver cancer. The research question we aim to answer is: *"What genes play the most important role in determining risk of hepatic cancer, and how can expression levels of these genes be used to inform diagnosis of liver cancer?"* We will apply data analysis techniques to the GSE263786 dataset to answer this question.

Data Preprocessing: To preprocess the data, we will load the data and the metadata from the URL dataset links from the NIH database website. We will have a dataframe of the data itself and a dataframe of the metadata. We will inspect the missingness to confirm that no patterns exist in the missing data, and then apply mean imputation to fill missing values. Mean imputation minimizes the loss of data. We will apply standard scaling to standardize the data. We will perform feature selection using principal component analysis (PCA); since PCA removes low-variance features while retaining specific amounts of variance in the dataset, no other feature selection techniques are necessary. The final data will be summarized in a correlation matrix and in histograms of top correlated features.

Unsupervised Learning: After preprocessing the data, principal component analysis (PCA) will be performed to reduce dimensionality while retaining 90% of the cumulative variance. This ensures that the most informative features are preserved while removing noise and redundancy from the dataset. The reduced data will then be visualized using scatter plots to reveal underlying structure and potential groupings within the data. Following this, K-means clustering will be applied to partition genes into distinct clusters based on their expression profiles. The optimal number of clusters will be determined using the elbow method, which identifies the point at which adding additional clusters yields diminishing returns. The resulting clusters will then be used to identify genes of interest for downstream network analysis.

Network Construction: Once clusters are identified, both a PPI network and a GRN network will be constructed. For the PPI network, the UniProt database will be connected via API. The PPI network will be built using the STRING algorithm, which assigns confidence scores to protein interactions and retains only those above a set threshold. For the GRN, the GenBank database will be connected via API. The GRN will be constructed using WGCNA, which groups highly co-expressed genes into modules and correlates them with phenotypic traits to identify biologically relevant gene clusters. Hub genes within significant modules will be identified and used to infer regulatory relationships, forming the final GRN. Both networks will be visualized using Cytoscape to highlight key genes and interactions.

Hub Genes: To identify hub genes within the dataset, we will count the number of connections for each gene in the GRN and corresponding protein in the PPI. We will consolidate these values into a score that computes the weight of each gene on liver cancer. We will then set a specific threshold number of minimum connections to select a gene as a Hub gene.

Final Solution: With the key genes for liver cancer identified via hub gene identification, this information can be used to identify risk of liver cancer. The hub gene expression levels will be used to train a linear discriminant analysis (LDA) model or logistic regression model to generate a classifier for hepatic cancer (the model with higher accuracy will be selected). We will then evaluate the efficacy of the selected model and analyze the p-values of each feature to validate our conclusions regarding Hub gene identification.